

Early warning from near-misses

Surfacing emerging safety risk early — from public data

One worked example, and an operating model to sustain it.

NASA Aviation Safety Reporting System (ASRS) · 80,047 reports · 2011–2025

The idea — and how the signal is read

- **Accidents are rare and *lagging*. Precursors — near-misses — are common, and they come *first*.** ASRS is a national, public repository of these voluntary reports — filed by pilots, controllers, and mechanics, and rich in free-text narrative.

An *early-warning system* is simply this: **detect precursors trending upward before the accident — from data already held.**

One detection engine (shared): each label becomes a monthly *share* (not raw counts, so COVID/awareness swings don't drive it); a series fires on **robust $z \geq 2$** vs its 24-mo baseline, **persisting 3+ months**, with taxonomy-change & seasonality guards.

The two views differ only in how a report is labeled:

1 · Structured tags

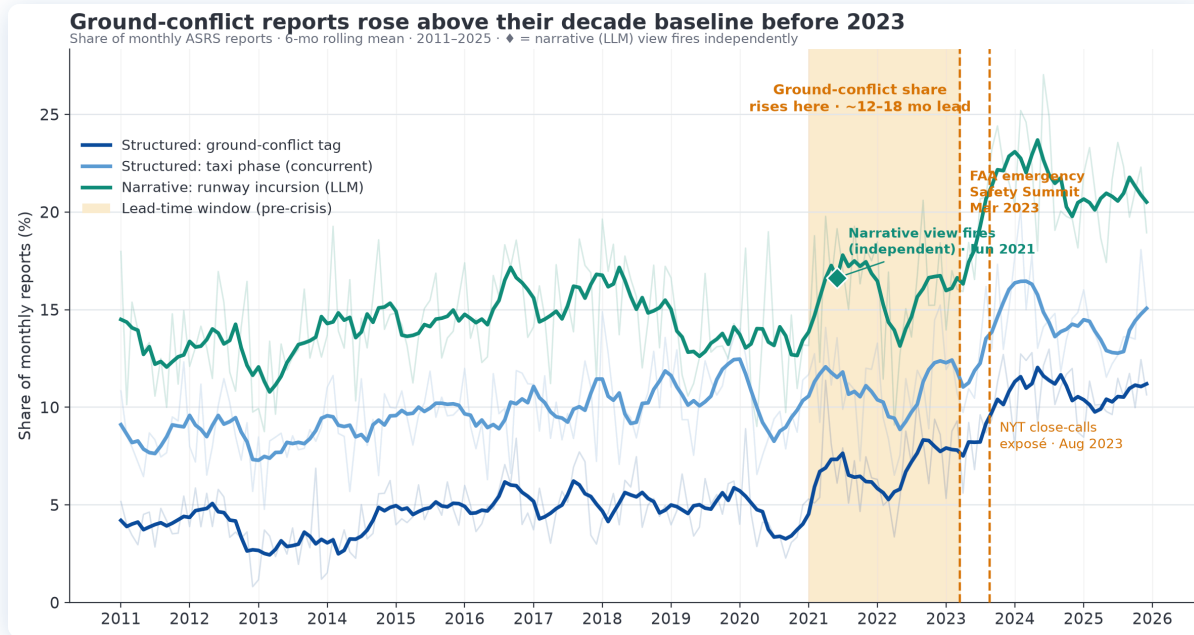
- From **ASRS taxonomy codes** — precise, but only risks that already carry a code.

2 · Narrative view (LLM)

- A **zero-shot model** reads the free text — catches the emerging & unlabeled (e.g. **GPS jamming**).

Output = a ranked shortlist for human review, not an alarm. Leading indicators, not lagging ones. · 79,572 narratives · 2011–2025 · US

Finding: runway & taxiway near-misses rose early



The structured "ground-conflict" tag roughly **doubled** — **~5% of reports to ~11% by 2024** (the raw count tripled too — not a denominator artifact). Independently, the narrative view (♦) began flagging runway-incursion reports in **June 2021** — both ahead of the 2023 crisis.

The telling part: the *structured* share spiked above threshold again and again (its z hit ~6) — but too erratically to ever hold the 3-month persistence rule, so it never registered a fire. The *narrative* signal rose smoothly enough to persist — and fired **June 2021. Same risk, two different failure modes — which is why both are run.**

It led the crisis — by 12–18 months

The signal rose in 2021–22. Public recognition came in 2023:

- **FAA emergency safety summit — March 2023**, after the JFK and Austin near-collisions.
- **NYT: airline close calls far more frequent than known — August 2023.**

→ **Lead time: ~12–18 months.**

The decisive check: the rise *predates* the summit — so awareness-driven over-reporting cannot explain the leading edge.

Notable: the FAA's own response was to *mine its safety database (ASIAS) for emerging trends* — **reactively, after** the events. Here the same idea is applied **beforehand**.

Not just more traffic — the *rate* rose too

The obvious objection: more flying ⇒ more conflicts.

The data rules it out:

- Incursions **per movement** ran **~15% above pre-pandemic** — rising while traffic was still **below** the 2019 peak.
- **Severity** spiked, not just frequency — and more traffic raises *frequency*, not *severity*.

Serious incursions (Cat A&B)	
2023	22 — a 6-year high
2024	9 (−59%)

Sources: FAA *Air Traffic by the Numbers*; DOT OIG (Mar 2025). External counts are fiscal-year, raw; the figures here are calendar-month share — directional, not like-for-like.

What this is — and what it isn't

What it is

- A *leading indicator* that risk is rising.
- Low-cost, public, explainable, reproducible.
- Corroborated two ways: the narrative detector that *fired*, the structured trend that *agrees*, and external sources.

What it isn't

- Not accident prediction — these are precursors, not accidents.
- Not causal. Leading hypothesis: post-COVID understaffing + the traffic rebound — but only the **symptom is observed, not the cause**.
- Voluntary data. The bias is *mitigated* (share, not counts) — not eliminated.
- Monthly, US-only, de-identified (no airport). These figures and the FAA's agree in **direction, not magnitude**.

Throughout: a signal is a reason for an analyst to *look* — never an automated verdict.

From one signal to a monthly watchlist

- This case is the **proof of concept** — the same pipeline already produces a **ranked monthly watchlist** from both detectors.
- **Value to a safety team:** a short, explainable list of *what is emerging and where to look first — months earlier*.

Monthly watchlist — ranked, explainable shortlist from both detectors

Candidates for human review, not alarms · 'peak z' = strength of the share anomaly · status from the persistence rule

Risk signal	View	First fire	Peak z	Status / note
GPS interference / jamming	Narrative	2019-05	27.0	watch
Runway / ground conflict	Both views	2021-06	4.2	watch · headline · structured trend agrees
Smoke / fire / fumes / odor	Both views	2019-05	5.3	watch
UAS / drone encounter	Narrative	2014-07	6.2	watch
Unstabilized approach	Narrative	2014-02	4.8	strong
Crew / ATC workload strain	Structured	2022-04	5.2	strong · post-COVID; reverting 2025

Next: join FAA traffic & controller-staffing data (to test causation) and NTSB outcomes (do these precursors lead to accidents?).

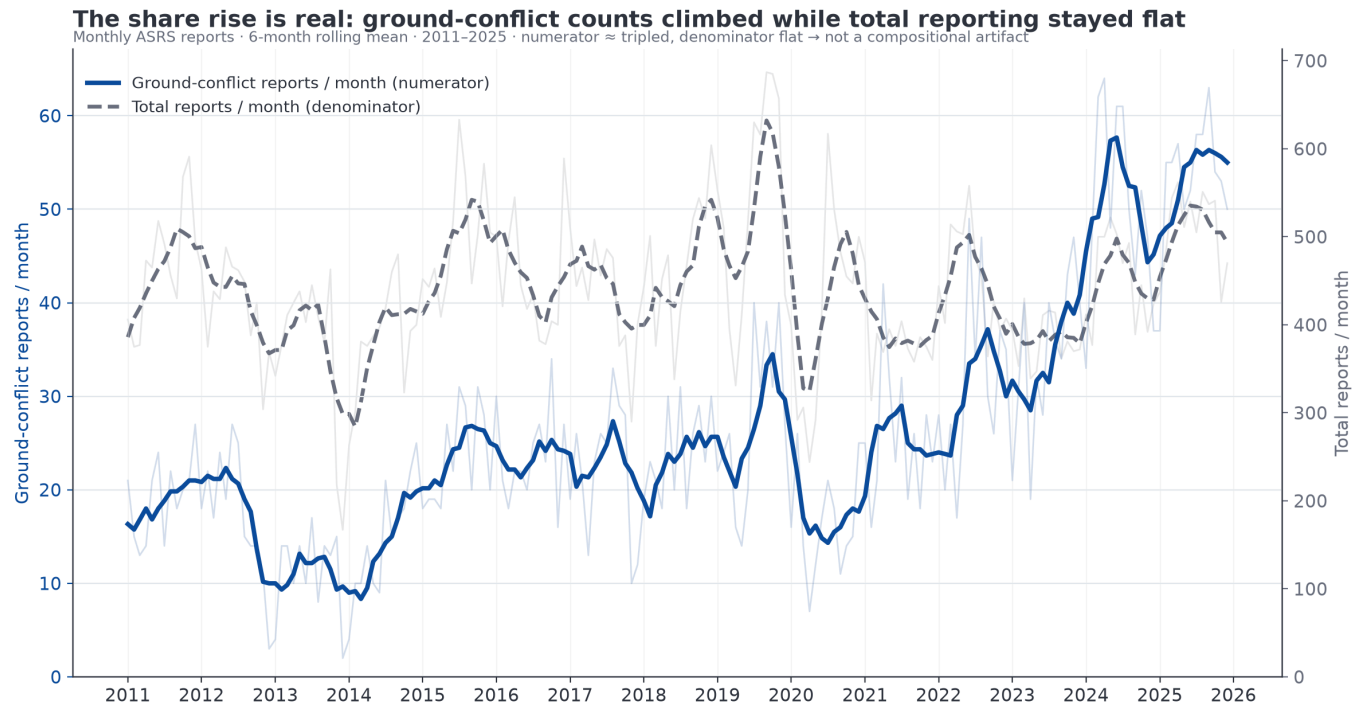
Thank you

Precursors come first — and they already sit in free, public data.

A simple, transparent method would have put runway conflict on an analyst's shortlist **~12–18 months early.**

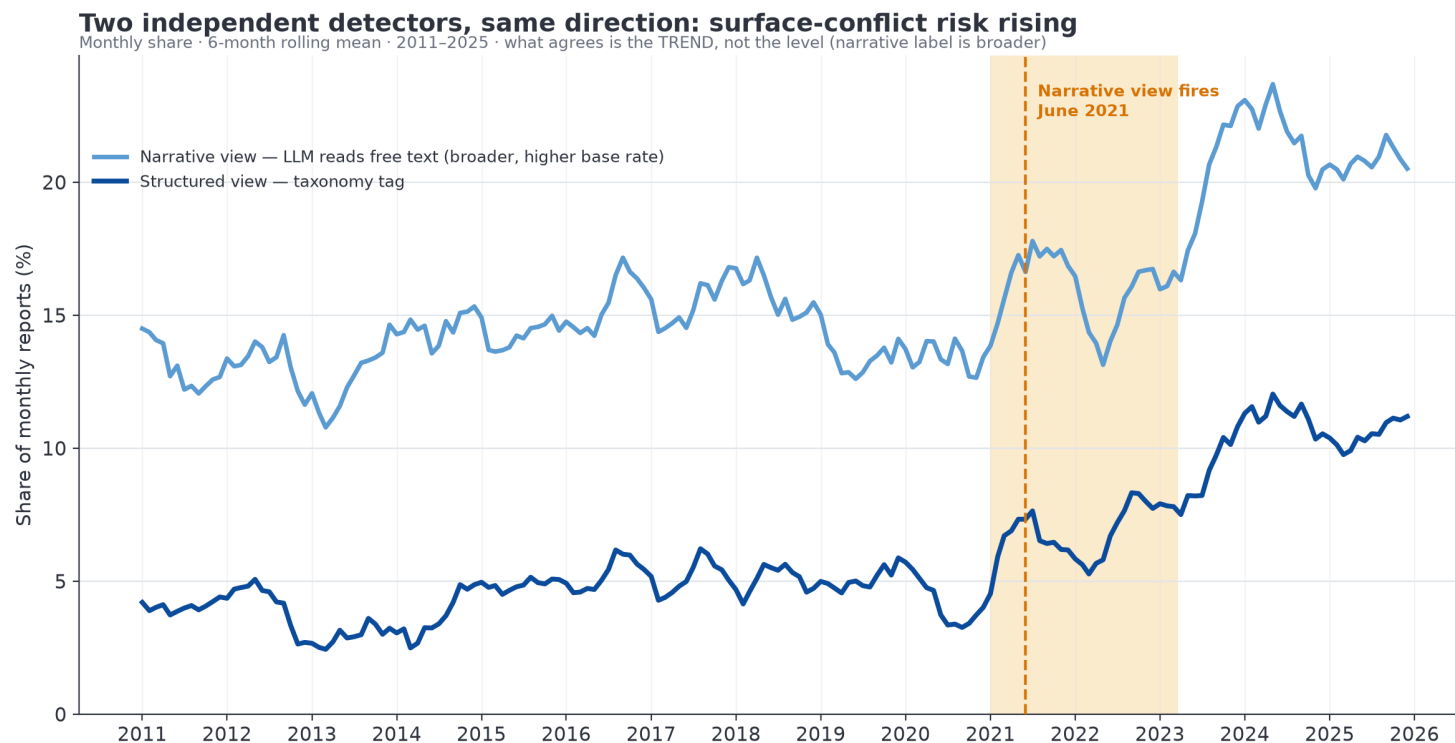
Full methodology & appendix available · ASRS 2011–2025 · narrative + structured views

Appendix — the share rise isn't a denominator artifact



A rising *share* could be an artifact if the denominator shrank or other categories collapsed. It isn't: the **raw ground-conflict count roughly tripled** ($\approx 230 \rightarrow 640$ reports/yr) while **total monthly reporting stayed flat** — the numerator rose on its own. ASRS anomaly tags are *multi-label*, so category shares don't sum to 100% (one rising doesn't force another down), and the FAA's *absolute* incursion counts rose independently. Share, raw count, and external data all point the same way.

Appendix — two independent detectors, same trend



The headline's cross-view check, on one axis. The **structured** taxonomy tag and the **narrative** LLM — entirely separate pipelines — both rise into 2021–24. What agrees is the *direction and timing*, not the level: the narrative label sits at a higher base rate because it's broader. The narrative view fires **June 2021**; the structured trend confirms. **(The 2022 share dip is a reporting-volume rebound — the underlying counts kept rising.)**

Appendix — was the language model validated? Yes

Reviewer spot-check — 30 flagged narratives (runway label):

- **70%** genuinely about surface / ground movement (broad match).
- **37%** a true runway-incursion / ground-conflict near-miss (strict).
- Precision rises with confidence: **88%** (7/8) at ≥ 0.95 · 70% (7/10) at 0.90 · 58% (7/12) at ≤ 0.85 — small bands, directional.
- **Main failure mode:** 8 of 9 false positives are *airborne* NMACs the model tags as "conflict."

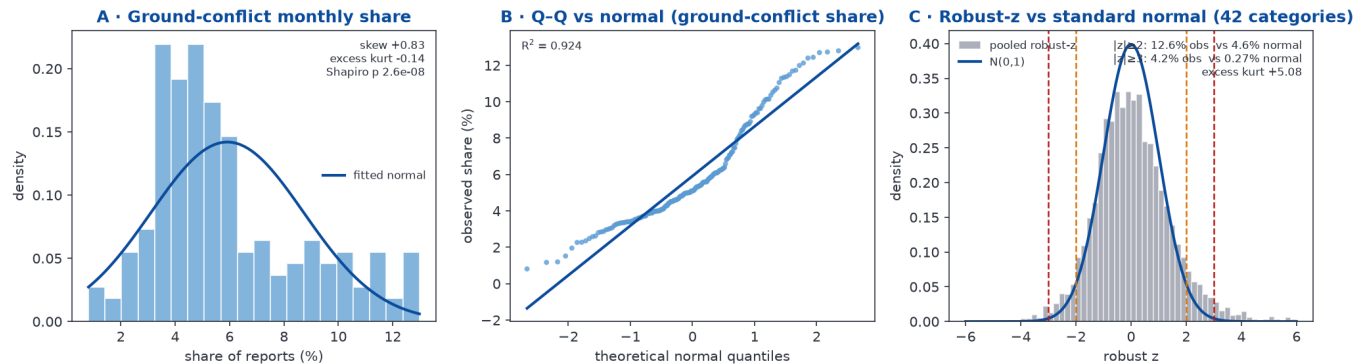
What it means — and the honest scope:

- The narrative *absolute level* is inflated, so **trend is used, not level**, cross-checked against the cleaner structured tag.
- A confidence threshold (≥ 0.95) or an airborne-vs-surface guard would sharpen it.
- Precision only (flagged positives); recall not measured — this is a spot-check, not a gold-standard eval.

Exactly why the output is a candidate for review, not a count.

Appendix — is the z-threshold a p-value? (distributional check)

Distributional diagnostic — the share series is non-normal; robust z is a screening rank, not a calibrated p-value



The robust z reads like a normal-theory score, so the distribution was checked. It **isn't normal** — the share series is right-skewed (Shapiro $p \approx 3e-8$) and the pooled robust-z is fat-tailed (excess kurtosis +5.1): **|z| ≥ 3 lands in 4.2% of category-months vs the 0.27% normality predicts (~15×)**. So median/MAD — which assume no distribution — keep the detector valid; the 2/3 cut-offs are a **screening rank, not calibrated p-values**; and the fat tails are exactly why **persistence, not a p-value, is the false-positive control**.